Q1 - 25 July - Shift 1

The sum of diameters of the circles that touch (i) the parabola $75x^2 = 64(5y - 3)$ at the point $\left(\frac{8}{5}, \frac{6}{5}\right)$ and (ii) the y-axis, is equal to _______.

Space for your notes:

Q2 - 25 July - Shift 2

The tangents at the point A(1, 3) and B(1, –1) on the parabola $y^2 - 2x - 2y = 1$ meet at the point P. Then the area (in unit^2) of the triangle PAB is :-

(A) 4
(B) 6
(C) 7
(D) 8

Space for your notes:

Q3 - 26 July - Shift 1

Let the function $f(x) = 2x^2 - \log_e x$, $x > 0$, be decreasing in (0, a) and increasing in (a, 4). A tangent to the parabola $y^2 = 4ax$ at a point P on it passes through the point (8a, 8a – 1) but does not pass through the point $\left(-\frac{1}{a}, 0\right)$. If the equation of the normal at P is $\frac{x}{\alpha} + \frac{y}{\beta} = 1$, then $\alpha + \beta$ is equal to-

Space for your notes:

Q4 - 26 July - Shift 2

Let P and Q be any points on the curves $(x-1)^2+(y+1)^2=1$ and $y = x^2$, respectively. The distance between P and Q is minimum for some value of the abscissa of P in the interval

(A) $\left(0,\frac{1}{4}\right)$ (B) $\left(\frac{1}{2},\frac{3}{4}\right)$

(C) $\left(\frac{1}{4},\frac{1}{2}\right)$ (D) $\left(\frac{3}{4},1\right)$

Space for your notes:

Q5 - 26 July - Shift 2

The equation of a common tangent to the parabolas $y = x^2$ and $y = -(x-2)^2$ is

(A) $y = 4(x-2)$ (B) $y = 4(x-1)$

(C) $y = 4(x+1)$ (D) $y = 4(x+2)$

Space for your notes:

Q6 - 27 July - Shift 1

Let P (a, b) be a point on the parabola $y^2 = 8x$ such that the tangent at P passes through the centre of the circle $x^2 + y^2 - 10x - 14y + 65 = 0$. Let A be the product of all possible values of a and B be the product of all possible values of b. Then the value of A + B is equal to :

(A) 0 (B) 25

(C) 40 (D) 65

Space for your notes:

Q7 - 27 July - Shift 2

If the length of the latus rectum of a parabola, whose focus is (a, a) and the tangent at its vertex is $x + y = a$, is 16, then $|a|$ is equal to :

(A) $2\sqrt{2}$ (B) $2\sqrt{3}$
(C) $4\sqrt{2}$ (D) 4

Space for your notes:

Q8 - 28 July - Shift 1

If the tangents drawn at the points P and Q on the parabola $y^2 = 2x - 3$ intersect at the point R(0, 1), then the orthocentre of the triangle PQR is :

(A) (0, 1) (B) (2, –1)
(C) (6, 3) (D) (2, 1)

Space for your notes:

Q9 - 28 July - Shift 2

Two tangent lines l_1 and l_2 are drawn from the point (2, 0) to the parabola $2y^2 = -x$. If the lines l_1 and l_2 are also tangent to the circle $(x - 5)^2 + y^2 = r$, then 17r is equal to

Space for your notes:

Answer Key

Q1 (10)	**Q2** (D)	**Q3** (45)	**Q4** (C)
Q5 (B)	**Q6** (D)	**Q7** (C)	**Q8** (B)
Q9 (9)			

Q1 (10)

$$x^2 = \frac{64.5}{75}\left(y - \frac{3}{5}\right)$$

equation of tangent at $\left(\frac{8}{5}, \frac{6}{5}\right)$

$$x \cdot \frac{8}{5} = \frac{64}{15}\left(\frac{y + \frac{6}{5}}{2} - \frac{3}{5}\right)$$

$$3x - 4y = 0$$

equation of family of circle is

$$\left(x - \frac{8}{5}\right)^2 + \left(y - \frac{6}{5}\right)^2 + \lambda(3x - 4y) = 0$$

It touches y axis so $f^2 = c$

$$x^2 + y^2 + x\left(3\lambda - \frac{16}{5}\right) + y\left(-4\lambda - \frac{12}{5}\right) + 4 = 0$$

$$\frac{\left(4\lambda + \frac{12}{5}\right)^2}{4} = 4$$

$$\lambda = \frac{2}{5} \text{ or } \lambda = -\frac{8}{5}$$

$$\lambda = \frac{2}{5}, \quad r = 1$$

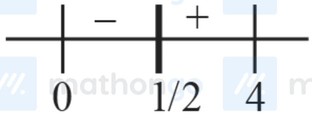
$$\lambda = -\frac{8}{5}, \quad r = 4$$

$$d_1 + d_2 = 10$$

Q2 (D)

(D)

Q3 (45)



$$f'(x) = 4x - \frac{1}{x}$$

$$a = \frac{1}{2}$$

Let $P(x_1, y_1)$ be any point on $y^2 = 4ax$

$$\frac{1}{y_1} = \frac{3 - y_1}{4 - x_1} \Rightarrow y_1^2 - 6y_1 + 8 = 0$$

$$y_1 = 2, 4$$

$\Rightarrow P(8, 4)$ as $P(2, 2)$ rejected

Equation of normal at P.

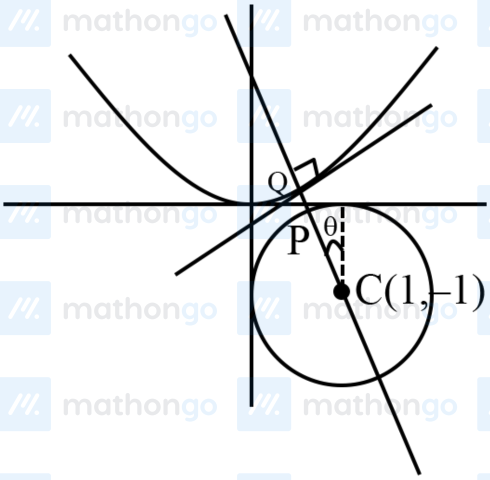
$$y - 4 = -4(x - 8)$$

$$\frac{x}{9} + \frac{y}{36} = 1$$

$$\alpha = 9, \beta = 36$$

$$\alpha + \beta = 45$$

Q4 (C)



$$Q=\left(t,t^{2}\right)$$

$$m_{CQ}=m_{normal}$$

$$\frac{t^{2}+1}{t-1}=-\frac{1}{2t}$$

Let $f(t)=2t^{3}+3t-1$

$$f\left(\frac{1}{4}\right)f\left(\frac{1}{3}\right)<0\Rightarrow t\in\left(\frac{1}{4},\frac{1}{3}\right)$$

$$P\equiv\left(1+\cos(90+\theta),-1+\sin(90+\theta)\right)$$

$$P=\left(1-\sin\theta,-1+\cos\theta\right)$$

$$m_{normal}=m_{CP}\Rightarrow-\frac{1}{2t}=\frac{\cos\theta}{-\sin\theta}\Rightarrow\tan\theta=2t$$

$$x=1-\sin\theta=1-\frac{2t}{\sqrt{1+4t^{2}}}=g(t)\quad(\text{let})$$

$$\Rightarrow g'(t)<0$$

$$g(t)\downarrow\text{function}$$

$$t\in\left(\frac{1}{4},\frac{1}{3}\right)$$

$$\Rightarrow g(t)\in(0.44,0.485)\in\left(\frac{1}{4},\frac{1}{2}\right)$$

Q5 (B)

Equation of tangent of $y = x^2$ be

$tx = y + at^2 \quad \ldots\ldots\ldots(1)$

$y = tx - \frac{t^2}{4}$

Solve with $y = -(x-2)^2$

$tx - \frac{t^2}{4} = -(x-2)^2$

$x^2 + x(t-4) - \frac{t^2}{4} + 4 = 0$

$D = 0$

$(t-4)^2 - 4 \cdot \left(4 - \frac{t^2}{4}\right) = 0$

$t^2 - 4t = 0$

$t = 0$ or $t = 4$

From eq. (1), required common tangent is

$y = 4(x-1)$

Q6 (D)

P(a, b) is point on $y^2 = 8x$, such that tangent at P pass through centre of $x^2 + y^2 - 10x - 14y + 65 = 0$

i.e. (5, 7)

Tangent at $P(at^2, 2at)$ is $ty = x + at^2$

A = 2 & it pass through (5, 7)

$7t = 5 + 2t^2$

$\Rightarrow t = 1,\ t = \frac{5}{2}$

$\therefore P\left(at^2, 2at\right) \Rightarrow (2,4)$ when t = 1

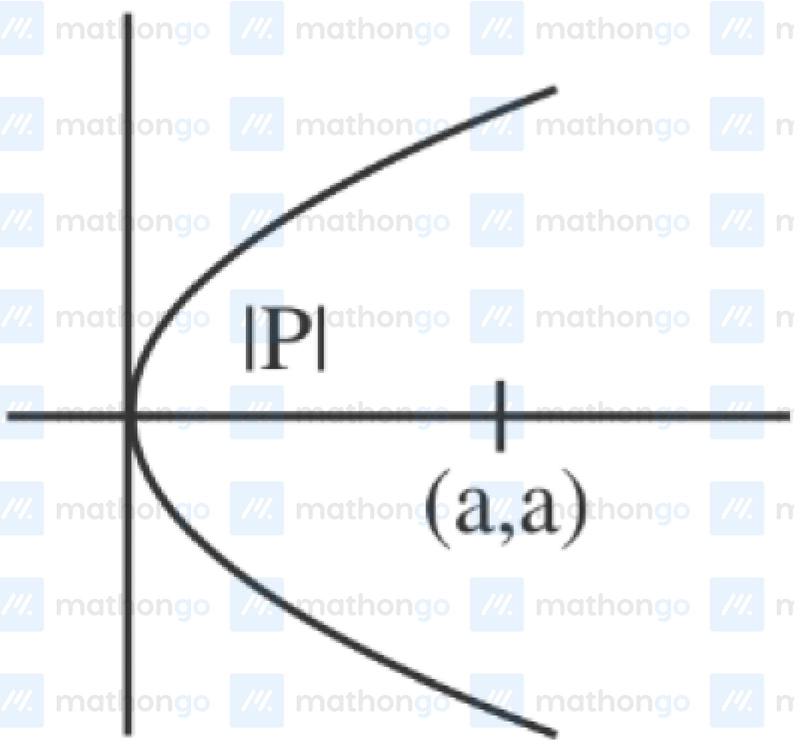
$\&\ \left(\frac{25}{2}, 10\right)$ when $t = \frac{5}{2}$

$\therefore A = 2 \times \frac{25}{2} = 25$

$B = 4 \times 10 = 40 \qquad \therefore A + B = 65$

Q7 (C)

$x+y=a$



$$|P|=\left|\frac{a}{\sqrt{2}}\right|=\frac{16}{4}=4$$

$$|a|=4\sqrt{2}$$

Ans. (C)

Q8 (B)

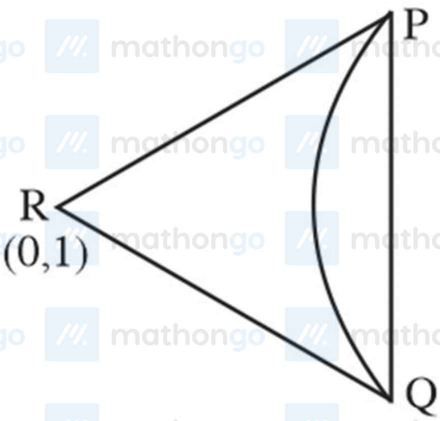
$$y^2=2x-3 \qquad \ldots(i)$$

Equation of chord of contact

$PQ: T=0$

$$y x 1 = (x + 0) - 3$$

$$y = x - 3 \qquad \ldots(2)$$



from (1) and (2)

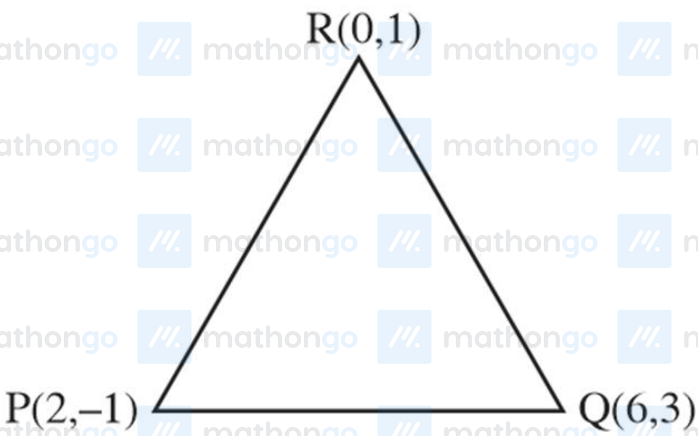
$$(x \cdot 3)^2 = 2x - 3$$

$$x^2 - 8x + 12 = 0$$

$$(x - 2)(x - 6) = 0$$

$$x = 2 \text{ or } 6$$

$$y = -1 \text{ or } 3$$



$$MP() = \frac{1}{} = 1$$

$$MQR = \frac{2}{6} = \frac{1}{3}$$

$$MPR = \frac{2}{6} = \frac{1}{3}$$

$$MPR = \frac{2}{-2} = -1$$

$$MPQ \times MPR = - \Rightarrow PQ \perp PR$$

Orthocentre = P (2,–1)

Q9 (9)

$$y^2 = -\frac{x}{2}$$

$$y = mx - \frac{1}{8m}$$

this tangent pass through (2, 0)

$m = \pm\frac{1}{4}$ i.e., one tangent is $x - 4y - 2 = 0$

$17r = 9$